

Sleep Health Prediction & Phenotyping in the All of Us Research Program

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SIADS 699 Capstone · University of Michigan School of Information · August 2026



Live dashboard



Slide deck

THE PROBLEM

Short and irregular sleep tracks cardiovascular disease, diabetes and depression. Wearables now record sleep at population scale, but who sleeps badly and how predictably is still mostly characterized from clinic samples and self-report.

58%

of this cohort averages fewer than seven hours a night — 26,348 people.

RESEARCH QUESTIONS

1. Prediction

How well do sociodemographic, health and activity features predict a person's sleep duration and consistency?

2. Phenotypes

What sleep-behavior phenotypes emerge from unsupervised clustering, and who is in them?

3. Fairness

Does that accuracy hold across racial, ethnic, and age subgroups?

THE COHORT

45,259

PARTICIPANTS

30.6M

NIGHTS

57

MEAN AGE

67%

FEMALE

6.89

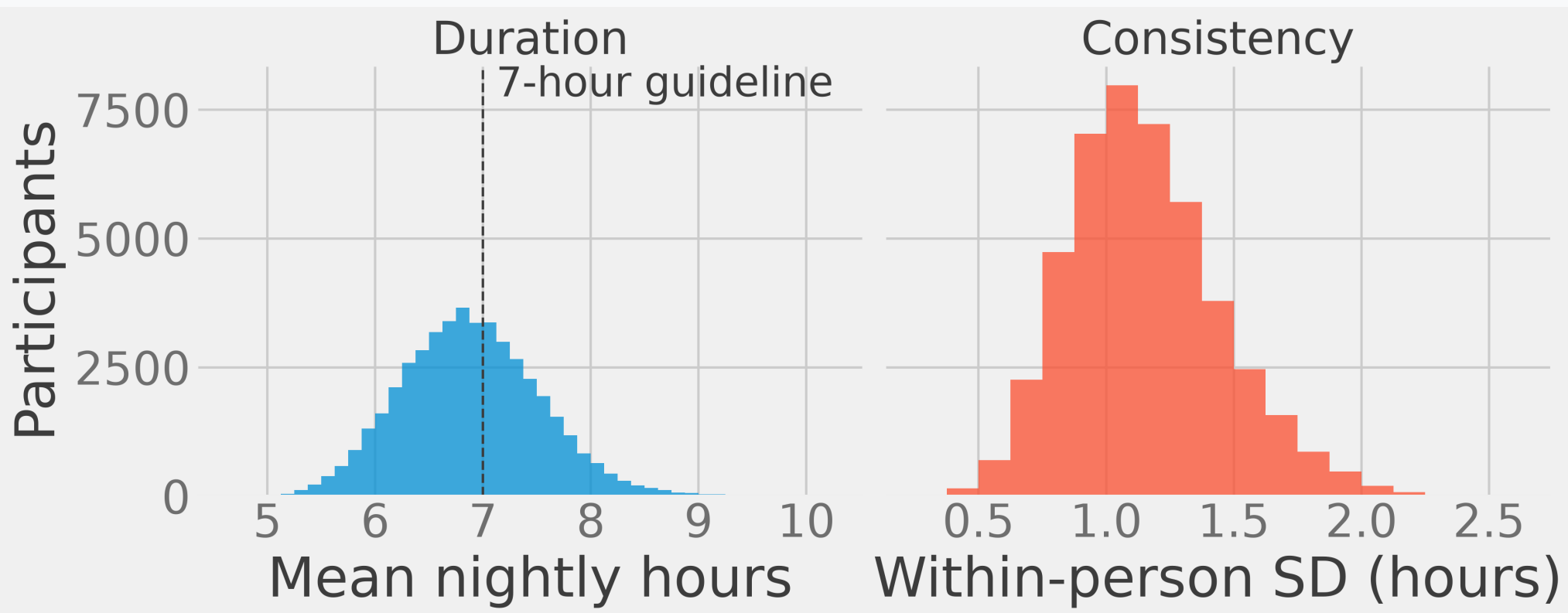
HOURS A NIGHT

7,081

DAILY STEPS

48,688 cleared four or more valid nights; 45,259 — 93% — cleared the 30-night floor. Below thirty nights, more than half the variance in the consistency target is estimator noise.

THE TWO TARGETS



One shared count axis and one bin width, so the heights compare directly: consistency is far more tightly concentrated than duration.

THE 17 FEATURES

- Numeric (9) — age, BMI, log(steps), steps CV, nights tracked, employment, education, self-rated health, income.
- One-hot (8) — gender (male, other) and race/ethnicity (Black, Hispanic or Latino, multiracial, other, unknown, White).
- Dropped — % short and % long nights and sleep IQR for leakage, the rest of the activity and wear-time blocks for collinearity, taking max |r| from 0.99 to 0.53 and the condition number from 24.5 to 5.2.

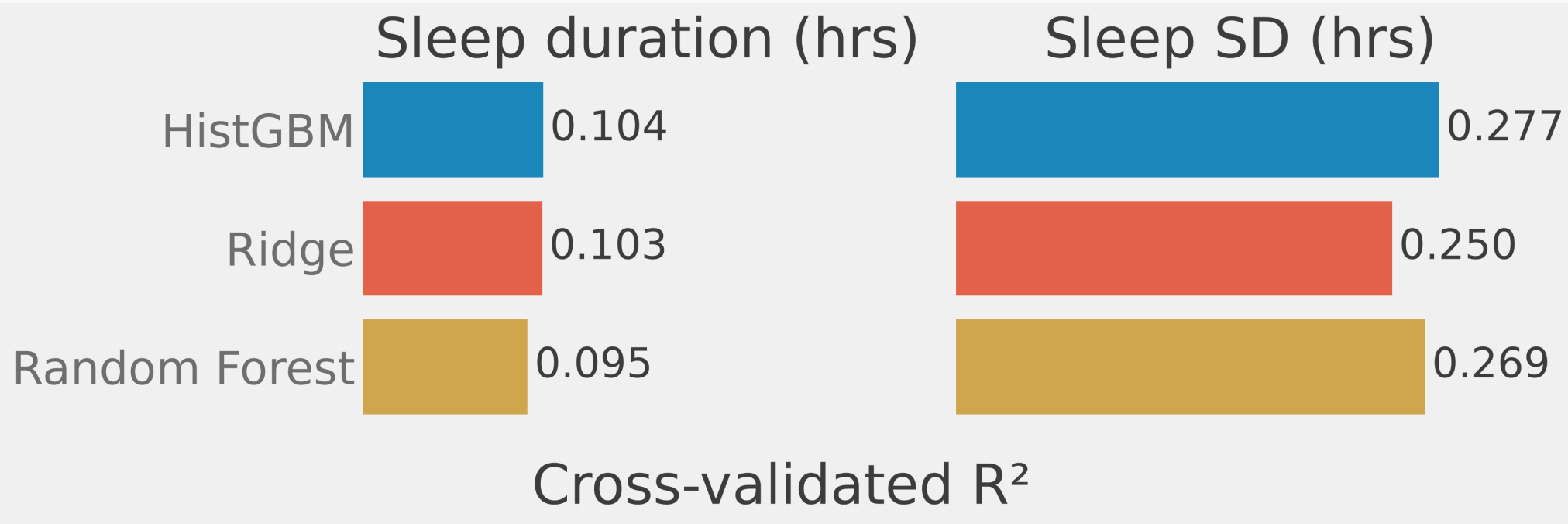
LIMITATIONS

- Fitbit owners in All of Us skew older, whiter and more female than US adults. Nothing here is nationally representative.
- Cross-sectional and observational — every association is correlational, and Fitbit staging is not a technical sleep study.
- No shift work, apnea, caffeine, or medication data, which is likely much of the duration signal we cannot reach.

R Q 1

Prediction

Four models, one feature matrix, five-fold cross-validation, so every difference below is the estimator, not the inputs.



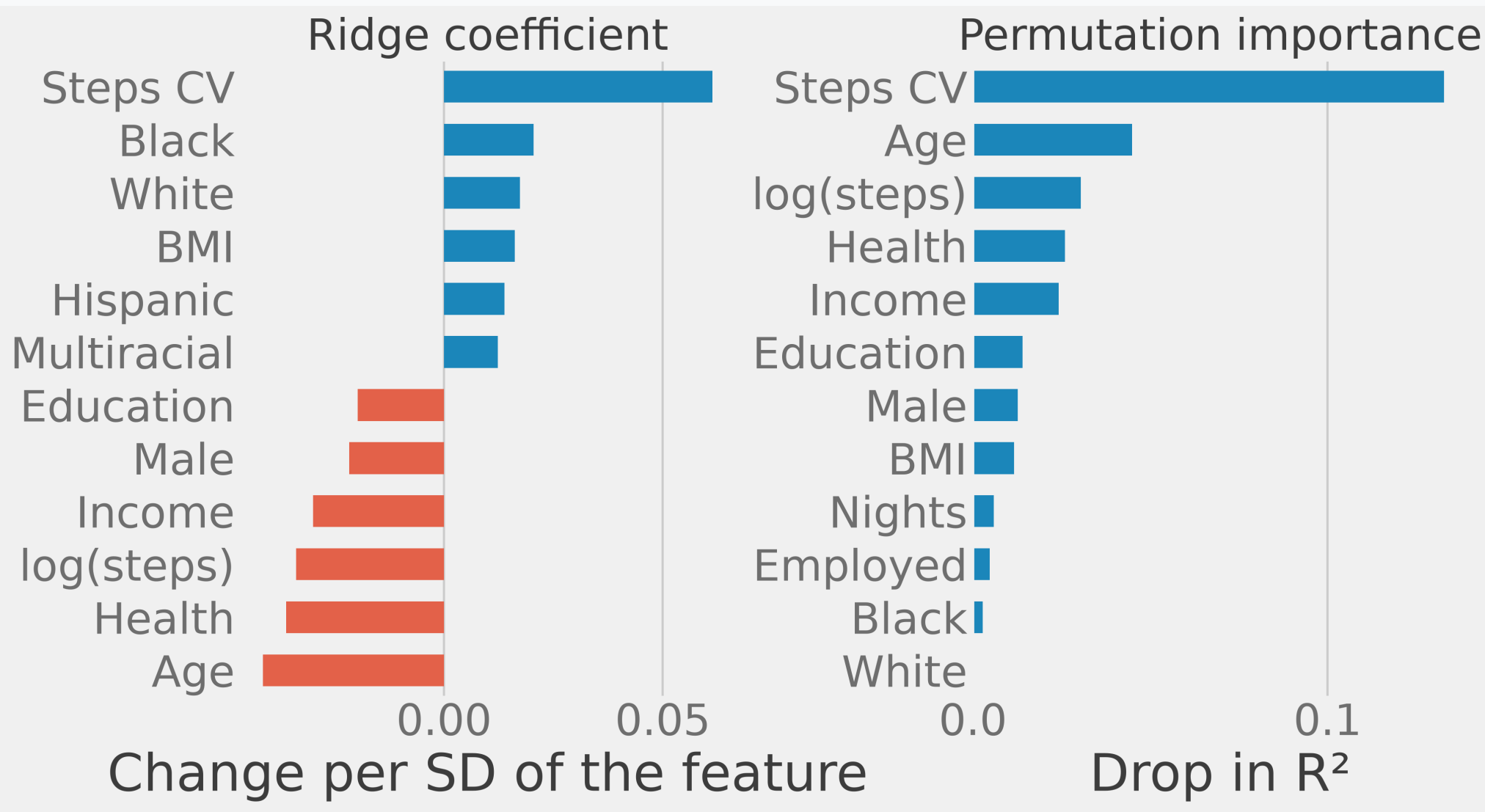
Both panels share one scale, so the bars are directly comparable. The mean baseline scores zero by construction and is not drawn.

Target	Model	R ²	RMSE	MAE
Consistency	HistGBM	0.277	0.252	0.198
	Random Forest	0.269	0.253	0.200
	Ridge	0.250	0.257	0.202
	Baseline (mean)	-0.000	0.296	0.235
Duration	HistGBM	0.104	0.612	0.482
	Random Forest	0.095	0.615	0.485
	Ridge	0.103	0.612	0.483
	Baseline (mean)	-0.000	0.646	0.513

R² = 0.28 is not a failed model

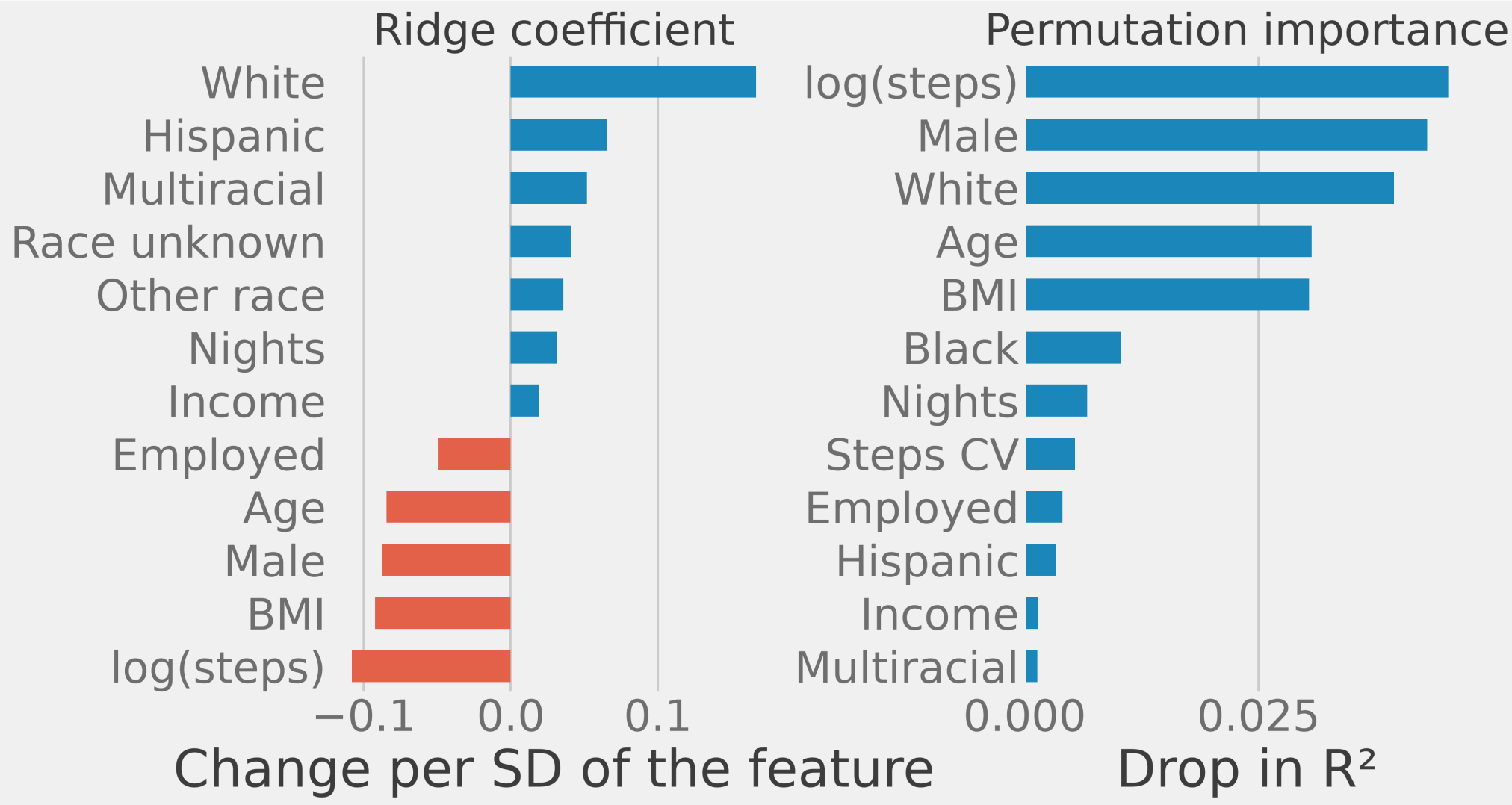
Individual sleep is dominated by things this data cannot see: shift work, apnea, caffeine, a newborn. The question is whether stable lifestyle traits carry signal at all, and against a mean baseline of zero they clearly do. On a held-out 20% every score lands at or above its cross-validated estimate, so none of this is selection optimism.

WHAT PREDICTS CONSISTENCY



Ridge coefficients on standardized features give the direction; HistGBM permutation importance gives the reliance.

WHAT PREDICTS DURATION



Nights tracked scores 0.007 — wear time is a control here, not a determinant.

R Q 2

Phenotypes

KMeans on four sleep-behavior columns. k = 4 is taken for the split it makes between regular and variable short sleepers.



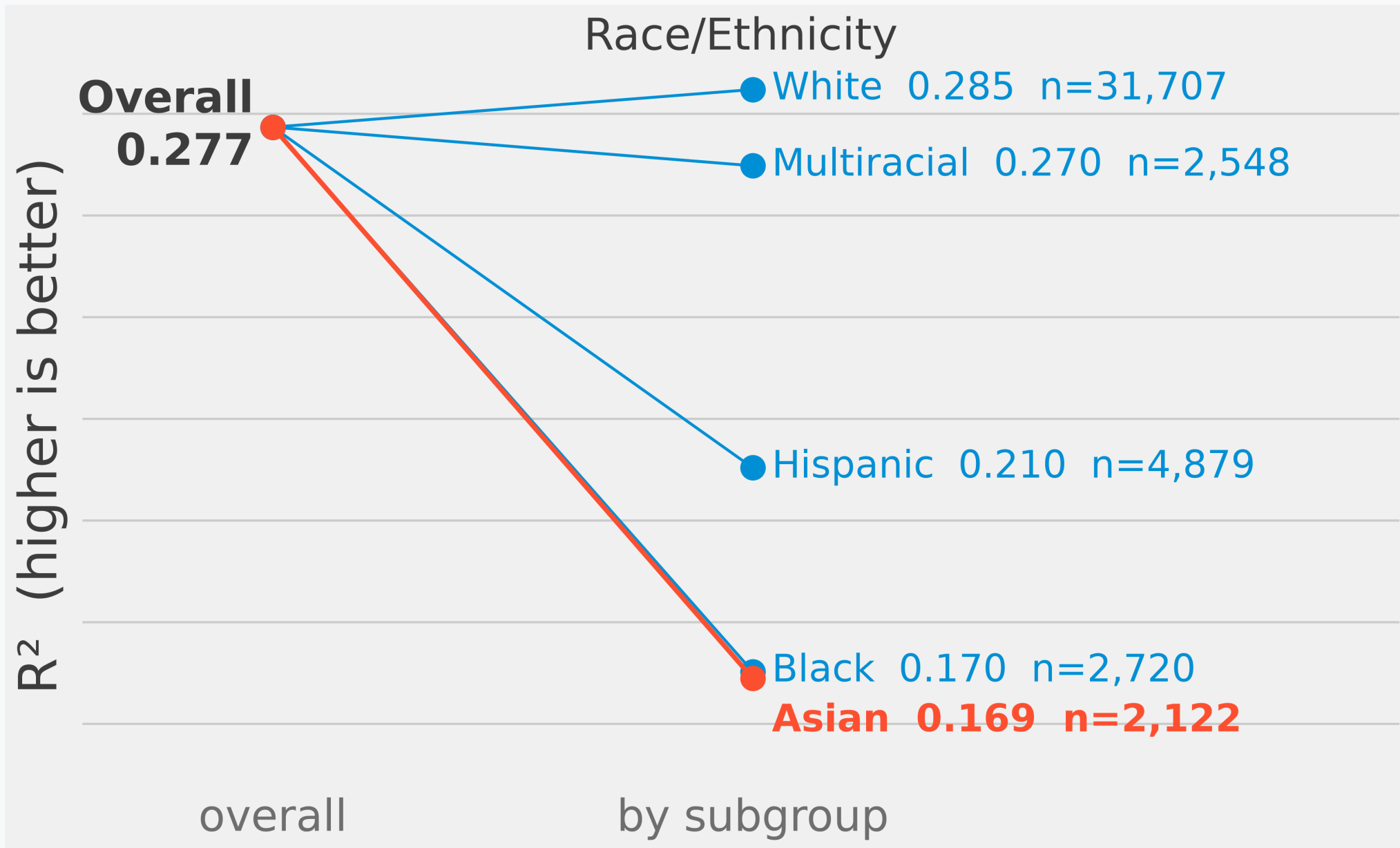
These are cuts through one continuous cloud, not natural clusters, but the cuts land where the health literature would put them.

R Q 3

Fairness

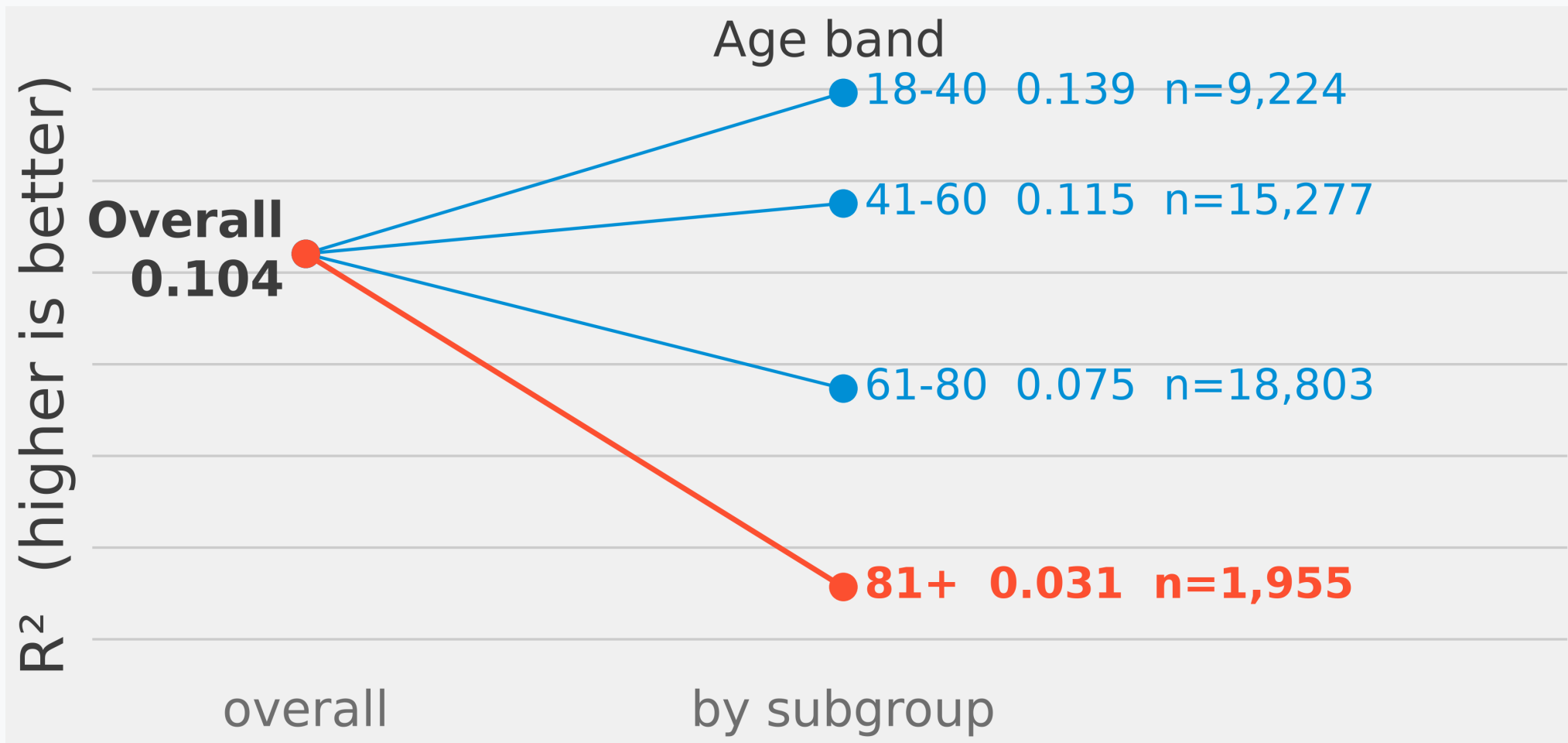
Out-of-fold R² computed within each subgroup, HistGBM, groups of at least n = 100.

Consistency, by race and ethnicity



Asian participants score 0.169 against 0.285 for White participants — 41% below. Each subgroup's R² uses its own mean as the denominator, so read across groups, not against the origin.

Duration, by age band



Accuracy falls 77% from 0.139 at 18-40 to 0.031 at 81+ — the largest single disparity here, on the target the models already predict least well.

What this means

0.28

R² on consistency, against 0.10 on duration. Lifestyle shapes how consistently people sleep far more than how long.

3x

Activity irregularity is the strongest single signal, about three times the next feature.

+11%

Model class matters only where there is signal: HistGBM beats Ridge on consistency, ties on duration.

Missing

What limits duration prediction is information the data does not carry, not the form of the model.

2 gaps

41% by race on consistency, 77% by age on duration. Accuracy is not evenly distributed, and the two do not share a pattern.